Name: Gaurang Vaghela
Rollno: BEAD-22561
CI Practical 1
Problem Statement: Design a distributed application using RPC for remote computation where client submits an integer value to the server and server calculates factorial and returns the result to the client program.

Server.py

```
from xmlrpc.server import SimpleXMLRPCServer
from xmlrpc.server import SimpleXMLRPCRequestHandler

class RPCServer:
    def calculate_factorial(self, n):
        if n < 0:
            raise ValueError("Input must be a non-negative integer.")
        result = 1
        for i in range(1, n + 1):
            result *= i
        return result

# Restrict to a particular path.
class RequestHandler(SimpleXMLRPCRequestHandler):
    rpc_paths = ('/api/v1/rpc_handler',)

# Create server
with SimpleXMLRPCServer(('localhost', 8000), requestHandler=RequestHandler) as server:
    server.register_introspection_functions()
    # Register the FactorialServer class"
    server.register_instance(RPCServer())
    print("FactorialServer is ready to accept requests.")
    # Run the server's main loop\n",
    server.serve_forever()
```

Client.py

```
import xmlrpc.client
# Create an XML-RPC client
with xmlrpc.client.ServerProxy("http://localhost:8000/api/v1/rpc_handler") as proxy:
    try:
        input_value = int(input("Enter number: "))
        result = proxy.calculate_factorial(input_value)
        print(f"Factorial of {input_value} is: {result}")

    except Exception as e:
        print(f"Error: {e}")
```

OUTPUT:

Server:

```
vboxuser@Ubuntu-24: ~/Desktop/CI1        vboxuser@Ubuntu-24: ~/Desktop/CI1
vboxuser@Ubuntu-24:~/Desktop/CI1$ python3 Server.py
FactorialServer is ready to accept requests.
127.0.0.1 - - [17/Mar/2026 09:34:34] "POST /api/v1/rpc_handler HTTP/1.1" 200 -
```

Client:

```
vboxuser@Ubuntu-24: ~/Desktop/CI1        vboxuser@Ubuntu-24: ~/Desktop/CI1
vboxuser@Ubuntu-24:~/Desktop/CI1$ python3 Client.py
Enter number: 5
Factorial of 5 is: 120
vboxuser@Ubuntu-24:~/Desktop/CI1$
```